

Introduction to classical encryption algorithms

Module II

Classical Ciphers

- Plaintext is viewed as a sequence of elements (e.g., bits or characters)
- **Substitution cipher:** replacing each element of the plaintext with another element. Substitution ciphers are classified as monoalphabetic and polyalphabetic ciphers.
- **Transposition (or permutation) cipher:** rearranging the order of the elements of the plaintext.
- **Product cipher:** using multiple stages of substitutions and transpositions

Substitution Ciphers

- A substitution cipher replaces one symbol with another.
- **Monoalphabetic ciphers:** In this kind of substitution ciphers, a character (or a symbol) in the plaintext is always changed to the same character in the ciphertext regardless of its position in the text. If the algorithm says letter A is changed to letter D, every letter A is changed to letter D. In other words, the relationship between letters in the plaintext and the ciphertext is one-to-one.
- Example: Plaintext : hello Ciphertext: KHOOR
- **Polyalphabetic ciphers:** In this kind of substitution ciphers, Each occurrence of a character may have different substitute. The relationship between letters in the plaintext and the ciphertext is one-to-many.
- Example: Plaintext : hello Ciphertext: ABNZF

Additive Cipher (shift cipher or Caesar cipher)

- The simplest monoalphabetic cipher is the additive cipher. It is also called as shift cipher or Caesar cipher.
- Mathematically, map letters to numbers:
a, b, c, ..., x, y, z
0, 1, 2, ..., 23, 24, 25
Each character is assigned an integer in Z_{26} . The secret key also in Z_{26} .
- Then the general Caesar cipher is:
$$c = E_k(p) = (p + k) \bmod 26$$
$$p = D_k(c) = (c - k) \bmod 26$$
- Can be generalized with any alphabet.

Additive Cipher (shift cipher or Caesar cipher)

- Example: Use additive cipher with key=15 to encrypt the message "hello".

Plaintext: h -> 07 Encryption: $(07+15) \bmod 26$ Ciphertext: 22 -> W

Plaintext: e -> 04 Encryption: $(04+15) \bmod 26$ Ciphertext: 19 -> T

Plaintext: l -> 11 Encryption: $(11+15) \bmod 26$ Ciphertext: 00 -> A

Plaintext: l -> 11 Encryption: $(11+15) \bmod 26$ Ciphertext: 00 -> A

Plaintext: o -> 14 Encryption: $(14+15) \bmod 26$ Ciphertext: 03 -> D

Additive Cipher (shift cipher or Caesar cipher)

- Example: Use additive cipher with key=15 to decrypt the message "WTAAD".

Ciphertext: W -> 22 Decryption: $(22-15) \bmod 26$ Plaintext: 07 -> h

Ciphertext: T -> 19 Decryption: $(19-15) \bmod 26$ Plaintext: 04 -> e

Ciphertext: A -> 00 Decryption: $(00-15) \bmod 26$ Plaintext: 11 -> l

Ciphertext: A -> 00 Decryption: $(00-15) \bmod 26$ Plaintext: 11 -> l

Ciphertext: D -> 03 Decryption: $(03-15) \bmod 26$ Plaintext: 14 -> o

Cryptanalysis

- Additive ciphers are vulnerable to ciphertext-only attacks using exhaustive key searches (brute-force method). The key domain of the additive cipher is very small. There are only 26 keys.
- Additive ciphers are also subject to statistical attacks. This is essentially true if the adversary has a long ciphertext. Adversary can use frequency of occurrence of characters for a particular language.
- Fig. 1 shows the frequency of English text.
- Sometimes, it is difficult to analyze a ciphertext based only on information about the frequency of a single letter. We need to know the occurrence of specific letter combinations such as digrams (2-letter combinations) and trigrams (three letter combinations) for English text.

English Letter Frequencies

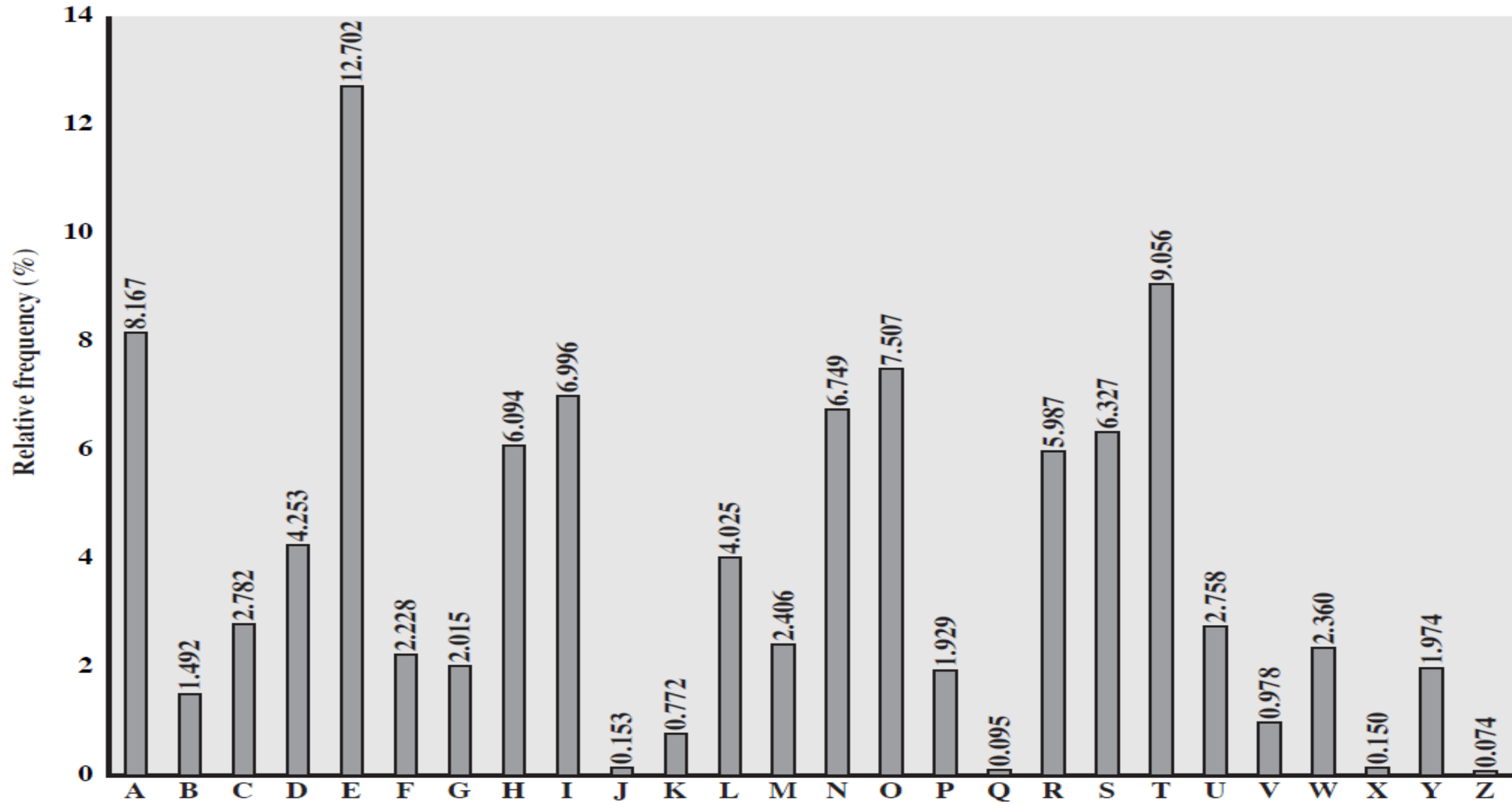


Fig. 1 Frequencies of alphabets of English language

Grouping of Digrams and Trigrams (Frequency based)

Diagram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF.
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH.

Multiplicative Ciphers

- The encryption algorithm specifies multiplication of a plain text by the key and the decryption algorithm specifies division of the ciphertext by the key.
- In multiplicative cipher, the plaintext and ciphertext are integers in Z_{26} ; the key is an integer in Z_{26}^*
- The key needs to be in Z_{26}^* . This set has only 12 members: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25. (numbers having multiplicative Inverses)

Multiplicative cipher

$$c = E_k(p) = (p \times k) \bmod 26$$

$$p = D_k(c) = (c \times k^{-1}) \bmod 26$$

Affine cipher

Affine cipher encryption

$$T = (p \times k1) \bmod 26$$

$$C = (T + k2) \bmod 26$$

Affine cipher decryption

$$P = (T \times k1^{-1}) \bmod 26$$

$$T = (C - k2) \bmod 26$$

- Combine the additive and multiplicative ciphers. The affine cipher uses a pair of keys in which the first key is from Z_{26}^* and the second is from Z_{26} . The size of the key domain is $26 \times 12 = 312$.

Monoalphabetic substitution cipher

- Since additive, multiplicative and affine ciphers have small key domains, they are very vulnerable to brute-force attack.
- A better solution is to create a mapping between each plaintext character and the corresponding ciphertext character.

a	b	c	d	e	f	g	h	i	j
N	O	T	A	R	B	E	C	F	U

k	l	m	n	o	p	q	r	s	t
X	D	Q	G	Y	L	K	H	V	I

u	v	w	x	y	z
J	M	P	Z	S	W

The size of the key space for the monoalphabetic substitution cipher is $26!$. This makes brute-force attack extremely difficult. However, statistical attack still can launch based on frequency of characters. Monoalphabetic ciphers do not change the Frequency of characters in the ciphertext, which makes the ciphers vulnerable to statistical attacks.

Poly alphabetic ciphers

In this ciphers, need to make each ciphertext character dependent on both the corresponding plaintext character and the position of the plaintext character in the message.

- **Autokey cipher :**

Cryptanalysis: Autokey cipher definitely hides the single letter frequency statistics of the plaintext. But it still vulnerable to brute-force attack.

$P = P_1 P_2 P_3 \dots$ $C = C_1 C_2 C_3 \dots$ $k = (k_1, P_1, P_2, \dots)$

Encryption $C_i = (P_i + k_i) \bmod 26$

Decryption $P_i = (C_i - k_i) \bmod 26$

Plain Text	a	t	t	a	c	k	i	s	t	o	d	a	y
P's value	00	19	19	00	02	10	08	18	19	14	03	00	24
Key stream	12	00	19	19	00	02	10	08	18	19	14	03	00
C's values	12	19	12	19	02	12	18	00	11	7	17	03	24
Cipher text	M	T	M	T	C	M	S	A	L	H	R	D	Y

Play fair cipher

- One approach to improving security is to **encrypt multiple letters at a time**.
- The **Playfair Cipher** is the best known such cipher.
- Invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair.
- Simplest substitution cipher with two letters combination.
- Encryption algo takes 5x5 matrix of letters.
- Generate the key table. (drop any duplicate letter).
- Key alphabets are filled in matrix from left to right & top to bottom.
- Rest of the letters are filled in matrix in remaining spaces.
- Letters I & j takes the same place.

Play fair cipher cntd..

- Rules:
 - If pair letters are same, add an X (uncommon letter) after the first letter.
 - Balloon will be (ba lx lo on).
 - If the letter appear in same row / column of the table, replace them with the letter to immediate right respectively.
 - If the letters are not on same row or column , replace with letter in the corners of rectangle.

Playfair Key Matrix

- Use a 5 x 5 matrix.
 - Fill in letters of the key (w/o duplicates).
 - Fill the rest of matrix with other letters.
 - E.g., key = **MONARCHY**.
- **Cryptanalysis:** The size of the key domain is 25!. But frequency of digram can be exploited to launch attacks.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example1 Playfair cipher

- Key: SCOPE, Plaintext: SECURITY, Ciphertext: CSEQQKYP
- Matrix:

S	C	O	P	E
A	B	D	F	G
H	I	K	L	M
N	Q	R	T	U
V	W	X	Y	Z

Example2 Playfair cipher

- Key: NETWORK, Plaintext: SECURITYLABVIT, Ciphertext: MORZCDWXPRKYGO
- Matrix:

N	E	T	W	O
R	K	A	B	C
D	F	G	H	I
L	M	P	Q	S
U	V	X	Y	Z

Hill Cipher

- The algo takes $n \times n$ matrix.
- The cipher C of P derived by multiplying P by K .
- When decrypt the message the inverse of K is used.
- $C = (KP) \bmod (26)$
- $P = K^{-1} C \bmod (26)$

Hill Cipher

- Example :-
 - Plaintext is “paymoremoney” and key is
 - $K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$
 - 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
 - **A B C D E F G H I J K L M N O P Q R S T**
 - 20 21 22 23 24 25
 - **U V W X Y Z**
 - **KEY PAY MOR EMO NEY**

Hill Cipher

- $PAY = \begin{bmatrix} 15 & 0 & 24 \end{bmatrix}$, $P = \begin{bmatrix} 15 \\ 0 \\ 24 \end{bmatrix}$
- $C = (KP) \bmod 26$

$$C = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix} \times \begin{bmatrix} 15 \\ 0 \\ 24 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 255+0+120 \\ 315+0+504 \\ 30+0+456 \end{bmatrix} \bmod 26$$

Hill Cipher

$$\bullet C = \begin{vmatrix} 3 & 7 & 5 \\ 8 & 1 & 9 \\ 4 & 8 & 6 \end{vmatrix} \pmod{26}$$

$$C = \begin{vmatrix} 11 & & L \\ 13 & & N \\ 18 & & S \end{vmatrix}$$

PAY = LNS

Vigenère Cipher

- Simplest polyalphabetic substitution cipher
- Consider the set of all Caesar ciphers:

$$\{ C_a, C_b, C_c, \dots, C_z \}$$

- Key: e.g. **security**
- Encrypt each letter using $C_s, C_e, C_c, C_u, C_r, C_i, C_t, C_y$ in turn.
- Repeat from start after C_y .
- Decryption simply works in reverse.

Example of Vigenère Cipher

- Keyword: *deceptive*

key: deceptive deceptive deceptive

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Security of Vigenère Ciphers

- There are multiple (how many?) ciphertext letters corresponding to each plaintext letter.
- So, letter frequencies are obscured but not totally lost.
- To break Vigenere cipher:
 1. Try to guess the key length. How?
 2. If key length is N , the cipher consists of N Caesar ciphers. Plaintext letters at positions k , $N+k$, $2N+k$, $3N+k$, etc., are encoded by the same cipher.
 3. Attack each individual cipher as before.

Transposition Ciphers

- Also called **permutation** ciphers.
- Shuffle the plaintext, without altering the actual letters used.
- Example: Row Transposition Ciphers

Keyed: Row Transposition Ciphers

- Plaintext is written row by row in a rectangle.
- Ciphertext: write out the **columns** in an order specified by a key.

Key: 3 4 2 1 5 6 7

Plaintext:

a	t	t	a	c	k	p
o	s	t	p	o	n	e
d	u	n	t	i	l	t
w	o	a	m	x	y	z

Ciphertext: **T****T****N****A****A****P****T****M****T****S****U****O****A****O****D****W****C****O****I****X****K****N****L****Y****P****E****T****Z**

Keyless Transposition ciphers

- Rail fence (method1):
- Plaintext is written zigzag fashion.

m e m a t e a k
e t e t h p r

- Ciphertext: first row followed by second row.

Ciphertext: memateaketethpr

Keyless Transposition ciphers

- Table method2:
- Plaintext is written in a table format

m	e	e	t
m	e	a	t
t	h	e	p
a	r	k	

- Ciphertext: column wise reading

Ciphertext: mmtaeehreaekttp

Keyed Transposition ciphers

- Plaintext : enemy attac kston ightz

3	1	4	5	2
1	2	3	4	5

- Ciphertext: EEMYN TAACT TKONS HITZG

Product Ciphers

- Uses a sequence of substitutions and transpositions
 - Harder to break than just substitutions or transpositions
- This is a bridge from classical to modern ciphers.

Unconditional & Computational Security

- A cipher is **unconditionally secure** if it is secure no matter how much resources (time, space) the attacker has.
- A cipher is **computationally secure** if the best algorithm for breaking it will require so much resources (e.g., 1000 years) that practically the cryptosystem is secure.
- All the ciphers we have examined are not unconditionally secure.

An unconditionally Secure Cipher

Vernam's one-time pad cipher

- Key = $k_1k_2k_3k_4\dots$ (random, **used one-time only**)
- Plaintext = $m_1m_2m_3m_4\dots$
- Ciphertext = $c_1c_2c_3c_4\dots$
where $c_i = m_i \oplus k_i$
- Can be proved to be unconditionally secure.